

Assignment

Section 3.1

1. Explain the OSI Model

Answer:

The OSI (Open Systems Interconnection) model is a framework used to understand how data is transmitted across a network. It divides communication into seven layers, each performing a specific function. Each layer interacts with the one above and below it to ensure smooth data transfer. It helps in troubleshooting and designing network systems.

Layer	Name	Example Protocol
7	Application	HTTP, FTP
6	Presentation	SSL/TLS
5	Session	NetBIOS
4	Transport	TCP/UDP
3	Network	IP
2	Data Link	MAC
1	Physical	Ethernet

TCP operates at the Transport Layer (Layer 4). It is connection-oriented and establishes a connection before sending data. It ensures reliable transmission using error checking, acknowledgments, and sequencing. It guarantees that data is delivered correctly and in order.

2. IP Addressing

Answer:

IP addressing is a method used to uniquely identify devices on a network. Each device is assigned an IP address for communication. It helps in routing data between devices over the internet or local network. IP addresses can be either public or private depending on their usage.

Public IP	Private IP
Globally unique	Used within local network
Assigned by ISP	Assigned by router
Accessible over internet	Not directly accessible
Less secure	More secure

Private IP ranges:

Class A: 10.0.0.0 – 10.255.255.255

Class B: 172.16.0.0 – 172.31.255.255

Class C: 192.168.0.0 – 192.168.255.255

192.168.1.100 is a private IP address because it falls within the Class C private range. It is used inside local networks and cannot be accessed directly from the internet.

3. Protocols

Answer:

Port	Service
22	SSH
80	HTTP
443	HTTPS
3306	MySQL

TCP	UDP
Connection-oriented	Connectionless
Reliable	Unreliable
Error checking	No error checking
Ordered delivery	No order guarantee
Slower	Faster
Used in web/email	Used in streaming/gaming

NAT (Network Address Translation) is used to convert private IP addresses into a public IP address. It allows multiple devices to share a single public IP. It modifies IP address information in packets as they pass through a router. It helps in conserving IP addresses and improving security. It also hides internal network structure.

NAT is needed because IPv4 addresses are limited. It allows efficient usage of available IP addresses.